

Final Assessment-Alternative Assessment

Semester	202520	Division	CIS
Assessment title in Syllabus	Final_Project_Portfolio	Program	Bachelor of Information Technology/ Bachelor of Information Systems

Course Code	CDS 2413		
Course Title	INTRODUCTION TO DATA SCIENCE		
Course Instructor		CRN	
Assessment	Project Deliverable 1 – Data Analysis Report	Submission Date	Week 6

Student Declaration:

Academic Integrity Statement

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إفادة النزاهة الأكاديمية

وفقاً لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبين وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض إلى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

Student HCT ID(s):	H00532802	H00533010		
Student Name(s):	Medeya Alhajri	Maryam AlHashmi		
Student Signature	Medeya	Maryam		

Chatbot (e.g. ChatGPT) statement

This submission allows you to use AI generated tools, but the prompt used must be stated and the response must be in red font. Include References to ChatGPT.

Instructions

Remove instructions and hints from the template before submitting the final report to BBLearn.

CLO1: Introduction and Data Wrangling (5 Points)

Purpose of Data Analysis (1 point)

The purpose of this data analysis is to study breast tumor data and understand how the characteristics of tumor cells are related to the diagnosis (malignant or benign).

The dataset contains measurements taken from microscope images of tumor cell nuclei. These measurements describe the size, shape, and structure of the cells. By analyzing this data, we want to discover patterns that help us predict whether a tumor is cancerous (malignant) or non-cancerous (benign).

The main goal is to use data analysis and machine learning techniques to build a model that can correctly classify tumors. This shows how data science can support medical diagnosis and help in early detection of breast cancer.

Type of Programming (1 point)

This project uses Python programming for data analysis. Python is chosen because it is easy to read and understand, which makes the coding process more efficient and organized. In addition, Python provides powerful libraries for data science, such as Pandas, NumPy, and Scikit-learn, which support data manipulation, analysis, and machine learning. Furthermore, Python is widely used in real-world data analytics across industries, making it a reliable and professional choice for this project.

The project follows a **step-by-step (procedural) approach**, which includes:

- 1-Loading the dataset
- 2-Cleaning the data
- 3-Analyzing the data
- 4-Creating graphs
- 5-Building machine learning models

The main libraries used are:

- **Pandas** for working with data tables
- **NumPy** for calculations
- **Matplotlib** for visualization
- **Scikit-learn** for machine learning

Python is suitable because it allows accurate, organized, and professional data analysis.

Type and Purpose of ML algorithm (1 point)

In our project, the dataset contains labeled data (diagnosis: M or B). Since the output is already known during training, the appropriate type is:

Supervised Learning

Supervised learning is used when the model is trained on labeled data to predict outcomes.

Within supervised learning, there are two main tasks:

- Regression (predict numerical values)
- Classification (predict categories)

Because our dependent variable (diagnosis) has only two categories:

- Malignant and Benign

This is a:

Binary Classification Problem

The purpose of using supervised classification is to build a model that can learn from existing tumor data and predict whether a new tumor is malignant or benign.

Identify the Variables (1 point)

Dependent Variable (Target Variable)

(Diagnosis)The variable **diagnosis** shows whether the tumor is:

- M = Malignant (cancerous)
- B = Benign (non-cancerous)

This is the dependent variable because it is the outcome we want to predict using the other features in the dataset.

It depends on the characteristics of the tumor cells. In other words, the values of radius, texture, area, concavity, and other measurements help determine whether the tumor is malignant or benign.

Since this variable is categorical (not numeric), it will need to be encoded into numbers (for example, 1 for Malignant and 0 for Benign) before applying classification algorithms.

Independent Variables (Predictor Variables)

The independent variables are the measurements taken from microscope images of tumor cell nuclei. These include:

Size-related features:

- Radius_mean
- Perimeter_mean
- area_mean

These variables measure how large the tumor cells are. Malignant tumors often have larger and more irregular cells.

Texture and smoothness features:

- Texture_mean
- Smoothness_mean

- These describe how uniform or rough the cells appear. Cancer cells usually look more irregular under a microscope.

Shape and irregularity features:

- Concavity_mean
- concave points_mean
- Symmetry_mean
- fractal_dimension_mean

- These measure how uneven or distorted the cell shape is. Malignant tumors often have more irregular shapes compared to benign tumors.

Additional Feature Groups:

- **Standard Error (_se) features**
These measure the variation or spread in the cell measurements.
- **Worst (_worst) features**
These represent the worst or most extreme measurements observed for each tumor.

All these variables are considered **independent** because they provide information that helps explain and predict the diagnosis.

Our dataset contains 30 independent variables. Each tumor measurement is recorded in three forms: mean (_mean), standard error (_se), and worst value (_worst). This means that for every main feature, there is an average value, a variation value, and an extreme value.

Sampling (1 point)

Sampling is the process of selecting a smaller group of data from a larger dataset for analysis. In our project, the dataset contains 569 records. Instead of analyzing the entire dataset for every statistical test, sampling is used to study a smaller subset and compare the results with the overall population.

Sampling is important because it saves time, reduces computational effort, and helps in performing statistical techniques such as hypothesis testing and t-tests. It also allows us to check whether a smaller sample can represent the full dataset accurately.

In our project, two sampling methods are used:

Simple Random Sampling

In this method, 150 records are selected randomly from the dataset. Each record has an equal chance of being chosen. This method reduces bias and ensures fairness in selection. Random sampling helps produce reliable and unbiased results.

Systematic Sampling

In systematic sampling, records are selected based on a fixed interval. For example, every k-th row from the dataset is chosen to create the sample. This method ensures that the sample is evenly distributed across the dataset and provides a structured way of selecting observations.

Both sampling techniques are applied to compare results and evaluate whether the sample represents the overall population effectively.

CLO2: Descriptive, Exploratory and Statistical Analysis (10 Points)

Purpose of Descriptive Analysis (1 point)

Descriptive analysis is performed to understand the basic structure, distribution, and behavior of the dataset before applying advanced statistical or machine learning techniques. Descriptive statistics summarize the dataset but do not allow us to draw conclusions about the population. Instead, they provide a clear overview of the data and prepare it for further analysis.

In our project, descriptive analysis includes measures of central tendency (mean, median, and mode), measures of dispersion (variance and standard deviation), measures of position (quartiles and skewness), and measures of frequency. These measures help us understand the overall distribution of tumor measurements and the diagnosis variable.

By performing descriptive analysis, we are able to identify patterns, detect possible outliers, examine variability in tumor characteristics, and check whether the dataset is balanced. This step ensures that the data is properly understood and suitable for sampling, visualization, hypothesis testing, and machine learning modeling.

Script for Descriptive Analysis (1 point)

A Python function is created to compute descriptive statistics for any numeric field in the dataset. This function calculates key measures such as **mean, median, mode, standard deviation, minimum, maximum, and quartiles**. Having a reusable function ensures consistency and efficiency when analyzing different samples or variables throughout the project.

```
[45]
✓ Os
import pandas as pd

df = pd.read_excel('breast cancer data.xlsx', header=1)

# Clean column names
df.columns = df.columns.str.strip()

# Encode diagnosis
df['diagnosis_num'] = df['diagnosis'].map({'M': 1, 'B': 0})

df.head()

def descriptive_stats(sample, field):
    s = sample[field].dropna()
    stats = {
        "count": s.count(),
        "mean": s.mean(),
        "median": s.median(),
        "mode": s.mode()[0] if not s.mode().empty else None,
        "std": s.std(),
        "min": s.min(),
        "25%": s.quantile(0.25),
        "50%": s.quantile(0.50),
        "75%": s.quantile(0.75),
        "max": s.max(),
        "range": s.max() - s.min()
    }
    return pd.DataFrame(stats, index=[field])
```

Random Sampling(1 point)

A simple random sample of **150 records** is selected from the dataset. Random sampling ensures that every record has an equal chance of being included, reducing selection bias. After selecting the sample, the descriptive statistics of the dependent variable **diagnosis_num** are calculated using the **descriptive statistics function**. This helps determine whether the sample reflects the overall distribution of benign and malignant cases in the population.

```
# Random sample of size 150
random_sample = df.sample(n=150, random_state=42)

# Descriptive statistics for diagnosis
random_desc = descriptive_stats(random_sample, 'diagnosis_num')
print(random_desc)
```

...	diagnosis_num	count	mean	median	mode	std	min	25%	50%	75%	\
		150	0.386667	0.0	0	0.488618	0	0.0	0.0	1.0	
		max	range								
	diagnosis_num	1	1								

Systematic Sampling (1 point)

Systematic sampling selects every k-th record from the dataset, ensuring that the sample is evenly distributed across the entire dataset. This method is useful when the dataset is large and uniformly structured. The descriptive statistics of the dependent variable are again calculated to compare how systematic sampling differs from random sampling. This comparison helps evaluate the representativeness and reliability of each sampling method.

The dataset has 569 records, so k was calculated as $569 \div 150 \approx 3$. Therefore, every 3rd record was selected to form the systematic sample.

```
# Determine interval
N = len(df)
k = N // 150

# Systematic sample
systematic_sample = df.iloc[::k].head(150)

# Descriptive statistics
systematic_desc = descriptive_stats(systematic_sample, 'diagnosis_num')
print(systematic_desc)
```

...	diagnosis_num	count	mean	median	mode	std	min	25%	50%	75%	\
		150	0.453333	0.0	0	0.499485	0	0.0	0.0	1.0	
		max	range								
	diagnosis_num	1	1								

Descriptive Statistical Report (1 point)

The dependent variable, **diagnosis_num** , is encoded as:

0 = Benign / 1 = Malignant

From the descriptive statistics:

- The **mean** indicates the proportion of malignant tumors in the dataset.
- The **median** and **mode** are typically 0, showing that benign tumors are more common.
- The **standard deviation** reflects the variability between benign and malignant cases.
- The **quartiles** confirm the imbalance in the dataset, with most values falling in the lower range.
- The **range** is always 1 because the variable is binary.

Summary:

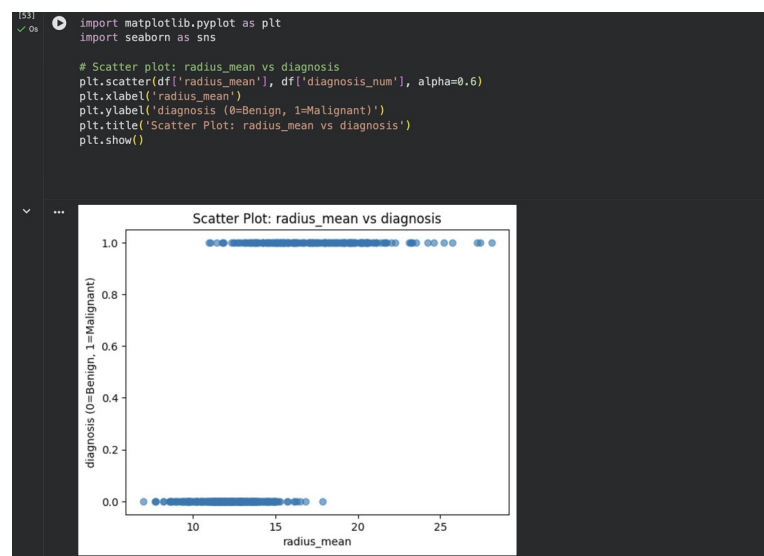
The dataset is imbalanced, with significantly more benign cases than others. This imbalance is important when developing machine learning models because it can influence how well the model performs, especially in classification tasks. To address this issue, techniques such as resampling the data or applying class weighting may be needed to improve model performance and ensure more balanced results.

Visualization through Various Charts (2 points)

The dependent variable is visualized using scatter plot, box plot, histogram, and heat map.

Scatter Plot :

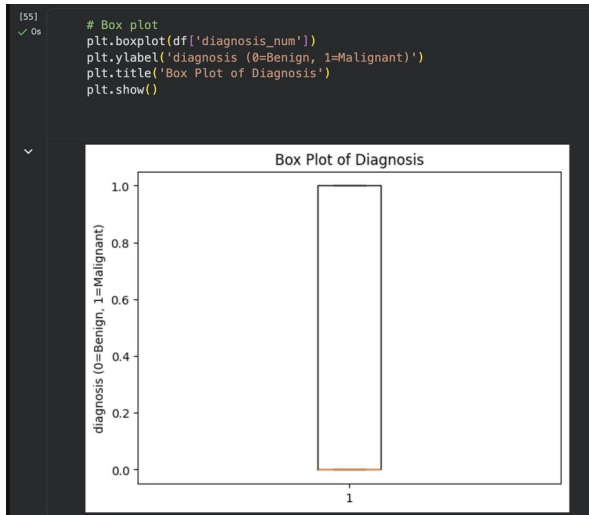
Shows the relationship between **radius_mean** and **diagnosis_num**. Larger tumor radius values tend to be associated with malignant diagnoses.



The scatter plot shows that larger tumor radius values are more common in malignant tumors. This indicates a positive relationship between tumor size and diagnosis.

Box Plot:

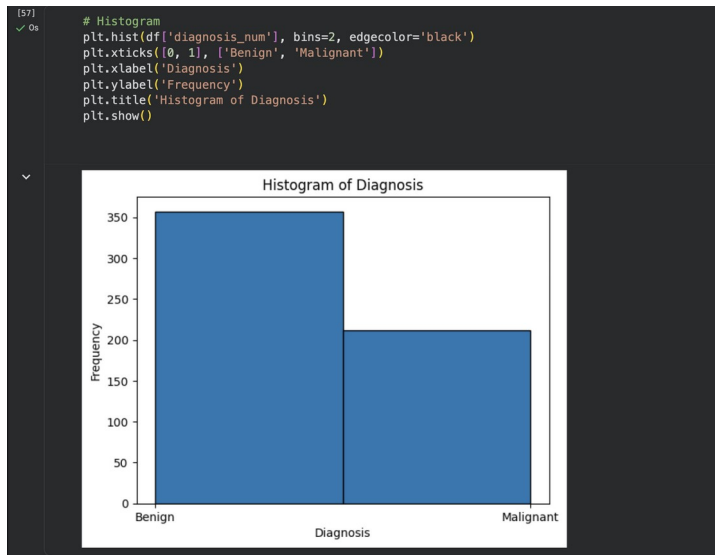
Displays the distribution of the diagnosis variable. The plot highlights the imbalance between benign and malignant cases.



The box plot confirms that the dataset is imbalanced, with more benign cases than malignant cases. This imbalance is important because it may affect the performance of machine learning models. Therefore, special attention may be needed during modeling to ensure accurate prediction of malignant cases.

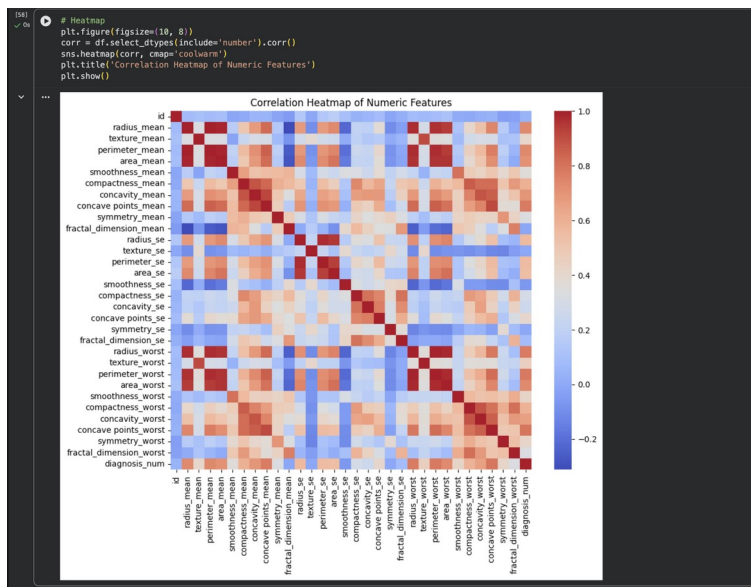
Histogram:

Shows the frequency of benign vs malignant tumors. The histogram clearly indicates that benign tumors are more common.



Heatmap:

Displays the correlation between all numeric variables. Strong correlations are observed between diagnosis and features such as **radius_mean**, **area_mean**, and **perimeter_mean**, suggesting these features are important predictors.



The heatmap confirms that tumor size-related features (radius, area, perimeter) are strong predictors of diagnosis. It also shows that some features are highly correlated with each other, which is important to consider during model building to avoid multicollinearity.

Overall, the heatmap supports the statistical test results and confirms that tumor size plays a significant role in predicting malignancy.

Hypothesis Test (1 point)

Hypothesis testing was performed to examine whether there is a meaningful relationship between tumor radius and diagnosis, tests are used:

Pearson Correlation (numeric vs numeric)

Measures the linear relationship between **radius_mean** and **diagnosis_num**.

```
[60]
✓ 0s from scipy.stats import pearsonr

pearson_r, pearson_p = pearsonr(df['radius_mean'], df['diagnosis_num'])
print(pearson_r, pearson_p)

... 0.7300285113754563 8.465940572264092e-96
```

In our dataset, the Pearson correlation showed a positive correlation between tumor radius and malignancy. The p-value was (8.47×10^{-96}) which is less than 0.05, indicating that the relationship is statistically significant. This suggests that larger tumor radius values are associated with malignant tumors.

- If $r > 0$, the relationship is positive.
- If $r < 0$, the relationship is negative.
- The closer $|r|$ is to 1, the stronger the relationship.

The p-value determines statistical significance:

- If $p < 0.05$, we reject the null hypothesis.
- If $p \geq 0.05$, we fail to reject the null hypothesis.

Spearman Correlation (rank-based)

Measures the rank-based relationship, useful when the data is not perfectly linear.

```
[62] from scipy.stats import spearmanr
spearman_r, spearman_p = spearmanr(df['radius_mean'], df['diagnosis_num'])
print(spearman_r, spearman_p)

... 0.7327849896210553 7.219245917007412e-97
```

The Spearman correlation test was used to check the relationship between `radius_mean` and `diagnosis_num`.

The Spearman correlation coefficient was **0.733**, which shows a strong positive relationship. This means that when the tumor radius increases, the chance of the tumor being malignant also increases.

The p-value was 7.22×10^{-97} , which is much smaller than 0.05.

Since the p-value is less than 0.05, we reject the null hypothesis and conclude that there is a statistically significant relationship between tumor radius and diagnosis.

This result confirms that tumor radius is strongly related to whether the tumor is malignant or benign.

Chi-Square Test (categorical vs categorical)

Tests the association between categorical variables by converting `radius_mean` into “High” and “Low” categories.

```
[64] import numpy as np
from scipy.stats import chi2_contingency

df['radius_cat'] = np.where(df['radius_mean'] > df['radius_mean'].mean(), 'High', 'Low')

contingency = pd.crosstab(df['radius_cat'], df['diagnosis'])
chi2, chi_p, dof, expected = chi2_contingency(contingency)
print(chi2, chi_p)

... 285.1548245576083 5.653357074691983e-64
```

For the Chi-Square test, the p-value was 5.65×10^{-64} , which is much smaller than 0.05. This shows a significant relationship between tumor size category (High/Low) and diagnosis.

One Sample T-test (2 points)

A one-sample t-test is used to check whether the random sample of 150 records represents the population in terms of diagnosis distribution.

```
[66] ✓ 0s from scipy.stats import ttest_1samp

population_mean = df['diagnosis_num'].mean()
sample_vals = random_sample['diagnosis_num']

t_stat, p_val = ttest_1samp(sample_vals, population_mean)
print(t_stat, p_val)

... 0.3530022105609316 0.7245856783929023
```

The p-value was **0.7246**, which is greater than 0.05. Therefore, there is no significant difference between the sample and the population, meaning the sample is representative of the population.

End of Documentation.